Does Spatial Correlation of Carbon Intensity Improve Robust Carbon-Aware Data-Center Scheduling?

A Multi-Grid Falsification Study with a Causal Mechanism — and a Phase-2 Copula Test

Marco Ortiz Togashi · Supervisor: Prof. Bissan Ghaddar · IE University — Master in Business Analytics & Data Science

Background & research questions

Data centers consume 1–2% of global electricity, and compute is unusually **flexible** — batch/training jobs can shift in time and across sites. Carbon intensity varies strongly by hour and region, so a scheduler can defer work to cleaner moments. Robust scheduling under *uncertain* carbon intensity uses **distributionally robust optimization (DRO)**. A natural extension treats carbon intensity as a stochastic *vector* so that **spatial correlation** can inform the schedule.

RQ1. Is carbon intensity of coupled grids genuinely correlated?
RQ2. Does feeding that correlation to a covariance-based DRO reduce out-of-sample tail emissions ($\mathrm{CVaR}_{0.95}$)?
RQ3. If not, *why* — and what dependence structure would a richer model need?

Method: a falsification test

A Mahalanobis–Wasserstein DRO, solved as a second-order cone program:

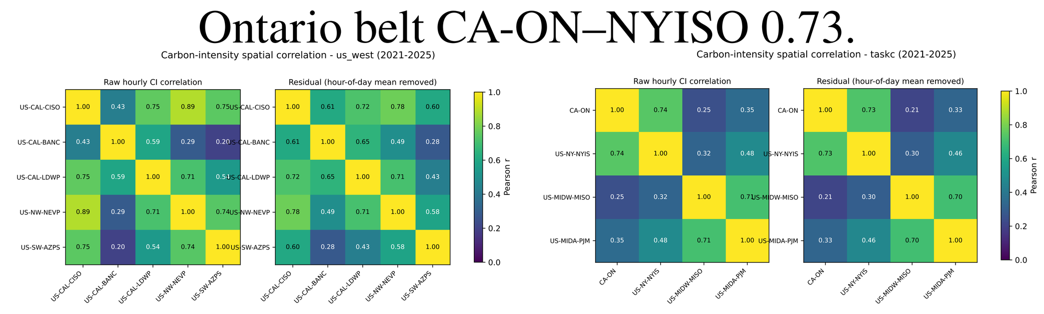
$$\min_{x\in X,\,t\geq 0}\ \langle \bar{\rho}, x\rangle + \varepsilon\, t \quad \text{s.t.} \quad \|L^\top \mathrm{vec}(x)\|_2 \leq t,\ \ LL^\top = \hat{\Sigma}.$$

Shuffled-marginals test. Fit the schedule twice — on the *joint* covariance vs. a *block-diagonal* one with all cross-region structure zeroed — and compare out-of-sample $\mathrm{CVaR}_{0.95}$ of daily emissions. The *spatial gap* is positive iff the joint covariance helps.

- Pre-registered: config commit-locked, single 2025 test read.
- Five region sets: 3 US/Canada grids spanning the dependence spectrum ($\bar{\tau} \approx 0 \to 0.5$) + an Iberia–France anchor.
- Blocked 5-fold CV selects $\varepsilon^\star$; 1000-bootstrap CIs.

RQ1 — the correlation is real

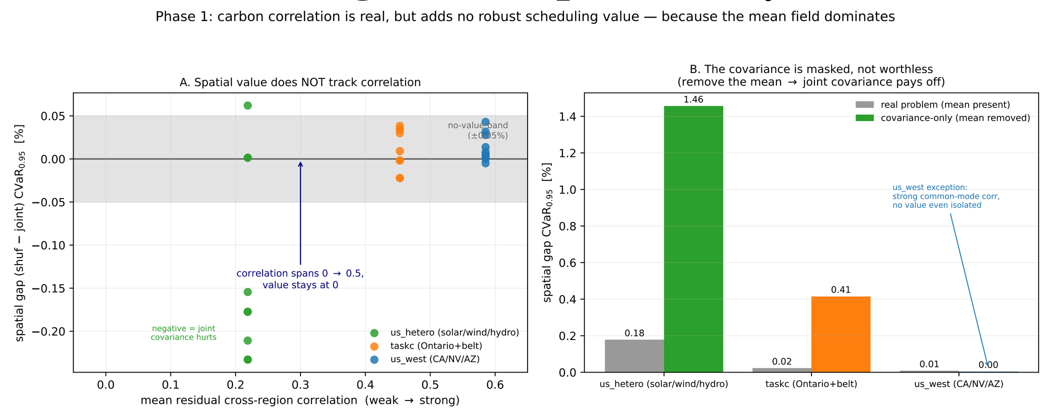
After removing each zone's hour-of-day mean, residual cross-region correlation is **strong and survives de-seasonalization**: US West CISO–NEVP 0.78; Ontario belt CA-ON–NYISO 0.73.



The assumption motivating spatial DRO is empirically valid.

RQ2 — but it adds no robust value

Across the dependence spectrum the spatial gap **never exceeds a few tenths of one percent** and is not sign-stable. It survives Ledoit–Wolf shrinkage, seasonal/AR(1) residualization, Benjamini–Hochberg correction over 144 cells, walk-forward to 2024, and a 3× **tighter-ramp** sensitivity.

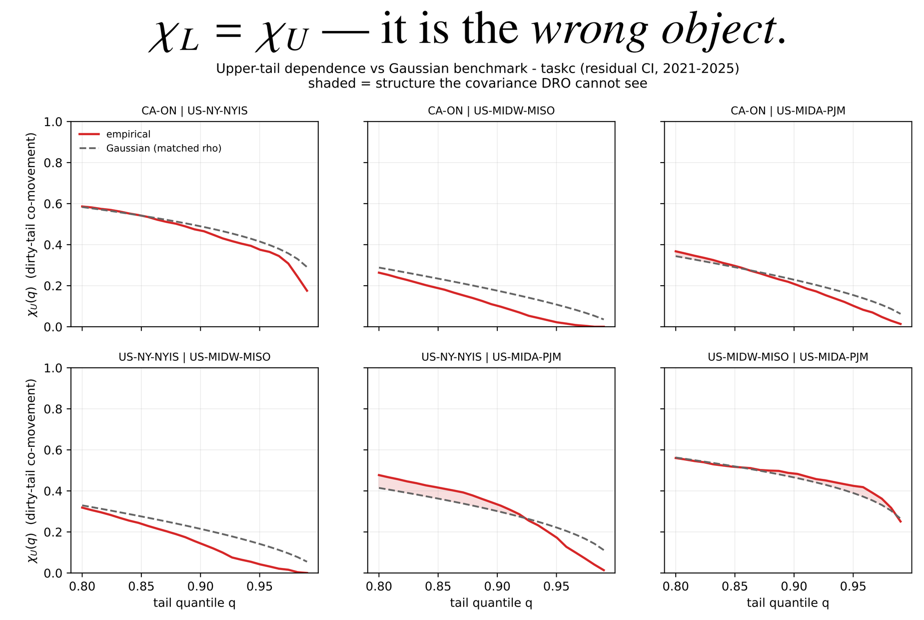


This is an **active null**: CV selects $\varepsilon^\star = 1$ in nearly every cell, so the DRO is genuinely robustifying — and the covariance *still* buys nothing.

RQ3 — the mechanism

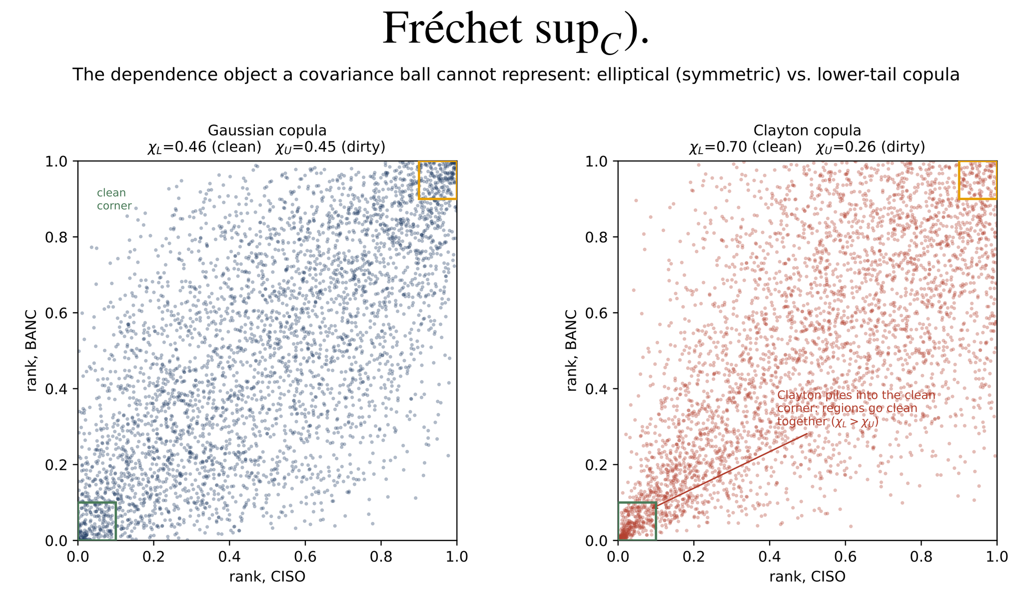
Mean-ablation (causal). Flatten the mean carbon field and the joint covariance suddenly pays — up to **+1.46%**. So the spatial signal is **real but masked** by the diurnal mean field, not absent.

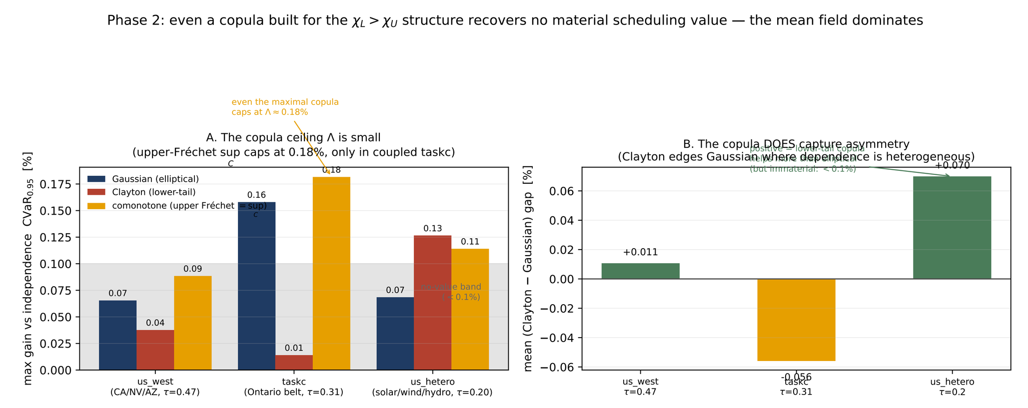
Tail dependence (structural). The residual dependence is **non-elliptical**: regions go *clean* together more than *dirty* together ($\chi_L > \chi_U$, upper-tail independent). An elliptical covariance ball forces $\chi_L = \chi_U$ — it is the *wrong object*.



Phase 2 — even the right copula fails

We replace the covariance ball with copula schedulers (CVaR sample-average) that *can* represent the asymmetry: independence, Gaussian, lower-tail **Clayton**, and the maximal **comonotone** (upper Fréchet $\sup_C$).





Even the *maximal* copula caps the gain at $\Lambda \approx$ **0.18%** — the null is **total**.

A mean-dominance bound

By translation invariance of CVaR, $\langle \rho, x\rangle = \langle \bar{\rho}, x\rangle + \langle \xi, x\rangle$, the dependence model moves only the residual term.

Proposition. For the SOCP, the spatial gap obeys $\left|\mathrm{OPT}(\Sigma) - \mathrm{OPT}(\Sigma^{\mathrm{shuf}})\right| \leq \varepsilon\,(\max_x \|x\|_2)\,\|\Sigma^{\mathrm{off}}\|_2^{1/2}$, vanishing as $\varepsilon \to 0$ or $\Sigma^{\mathrm{off}} \to 0$.

The tight ceiling is *measured* by the comonotone arm ($\Lambda \approx 0.18\%$) — an order below the mean-driven value.

Conclusions & future work

- Spatial correlation of carbon intensity is **real but adds no robust scheduling value** — anywhere on the spectrum, for *any* passive dependence model.
- The binding constraint is the **mean field**, not the shape of the dependence (mean-dominance bound + comonotone ceiling).
- **Recommendation:** per-region marginal schedulers capture essentially all the value at a fraction of the cost.
- **Next:** spatial value, if any, requires an *active* inter-region transfer channel $f_{r\to s,t}$ — the one route the bound does not close.

Reproducible pipeline: 169 unit tests, pre-registered configs, license-safe archived results. *AI used as a coding/drafting assistant under the author's direction.*

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